

PARTIAL PROOF TERMS IN THE STUDY OF PROOF SEARCH

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Finished Proofs

$$\frac{\frac{\frac{}{A, B \vdash A} Ax}{A \vdash B \supset A} I \supset}{\vdash A \supset B \supset A} I \supset$$



C.H. Isomorphism

Proof Terms

$$\lambda x^A. \lambda y^B. x$$

Finished Proofs

$$\frac{\frac{\frac{}{A, B \vdash A} Ax}{A \vdash B \supset A} I \supset}{\vdash A \supset B \supset A} I \supset$$

$\longrightarrow$

C.H. Isomorphism

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$$\lambda x^A. \lambda y^B. x$$

Unfinished Proofs

$$\frac{\frac{?}{A \vdash B \supset A}}{\vdash A \supset B \supset A} I \supset$$

$\longrightarrow$

?

Finished Proofs

$$\frac{\frac{\frac{}{A, B \vdash A} Ax}{A \vdash B \supset A} I \supset}{\vdash A \supset B \supset A} I \supset$$



C.H. Isomorphism

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In the literature...

Finished Proofs

$$\frac{\frac{\frac{}{A, B \vdash A} Ax}{A \vdash B \supset A} I \supset}{\vdash A \supset B \supset A} I \supset$$



C.H. Isomorphism

Proof Terms

$$\lambda x^A. \lambda y^B. x$$

Unfinished Proofs

$$\frac{?}{\frac{A \vdash B \supset A}{\vdash A \supset B \supset A} I \supset}$$



In the literature...

Meta-variables

e.g. [Muñoz, 2001]

- Model, with the help of explicit substitutions, the reuse of proofs/ delay of choices in unification and proof search;
- Used in the meta-theory of proof assistants based on type theory.

Finished Proofs

$$\frac{\frac{\frac{}{A, B \vdash A} Ax}{A \vdash B \supset A} I \supset}{\vdash A \supset B \supset A} I \supset$$



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Unfinished Proofs

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In the literature...

Open Proofs and Open Proof Terms

[Geuvers, Jojgov, 2002]

- Developed directly for HOL; deal with proofs in ND, possibly containing a generic "claim" inference;
- Does not have an explicit formalization of a search process.

Finished Proofs

$$\frac{\frac{\frac{}{A, B \vdash A} Ax}{A \vdash B \supset A} I \supset}{\vdash A \supset B \supset A} I \supset$$



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In the literature...

Coinductive Proof Search

e.g. [JES, Matthes, Pinto, 2021]

- Employs proof terms with meta-variables X^σ , allowing the representation of an entire search space;
- Does not yet provide means for expressing proof search procedures.

Finished Proofs

$$\frac{\frac{\frac{}{A, B \vdash A} Ax}{A \vdash B \supset A} I \supset}{\vdash A \supset B \supset A} I \supset$$



C.H. Isomorphism

Proof Terms

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Unfinished Proofs

$$\frac{\frac{?}{A \vdash B \supset A}}{\vdash A \supset B \supset A} I \supset$$



In the literature...

L-Framework

[Olarite, Pimentel, Rocha, 2023]

- Term rewriting as a meta-logical framework for the specification of sequent calculi and the automation of meta-theory;
- More focused in automation and bottom-up proof search.

Finished Proofs

$$\frac{\frac{\frac{}{A, B \vdash A} Ax}{A \vdash B \supset A} I \supset}{\vdash A \supset B \supset A} I \supset$$



C.H. Isomorphism

Proof Terms

$$\lambda x^A. \lambda y^B. x$$

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$$\frac{\frac{?}{A \vdash B \supset A}}{\vdash A \supset B \supset A} I \supset$$



Our approach...

Finished Proofs

$$\frac{\frac{\frac{}{A, B \vdash A} Ax}{A \vdash B \supset A} I \supset}{\vdash A \supset B \supset A} I \supset$$



C.H. Isomorphism

Proof Terms

$$\lambda x^A. \lambda y^B. x$$

Unfinished Proofs

$$\frac{\frac{?}{A \vdash B \supset A}}{\vdash A \supset B \supset A} I \supset$$



Our approach...
Extension of the
C.H. Isomorphism

Partial Proof Terms

$$\lambda x^A. (A \vdash B \supset A)$$

- *Formal sequents*, i.e., sequents occurring as components of proof terms;
- ***Idealized proof search*** - theoretical study of proof search without considerations of control, effectiveness or implementation.

A first case study: LJT and Focusing

Formulas

$$A, B ::= p \mid A \supset B$$

A first case study: LJT and Focusing

Formulas

$$A, B ::= p \mid A \supset B$$

Sequents

$$\Gamma \vdash A \qquad \Gamma; A \vdash p$$

A first case study: LJT and Focusing

Formulas

$$A, B ::= p \mid A \supset B$$

Sequents

$$\Gamma \vdash A \qquad \Gamma; A \vdash p$$

Inference Rules

$$\frac{}{\Gamma; p \vdash p} \text{ AX}$$

$$\frac{\Gamma, x : A; A \vdash p}{\Gamma, x : A \vdash p} \text{ CTR}$$

$$\frac{\Gamma, x : A \vdash B}{\Gamma \vdash A \supset B} \text{ RI}$$

$$\frac{\Gamma \vdash A \quad \Gamma; B \vdash p}{\Gamma; A \supset B \vdash p} \text{ LI}$$

Proof Terms and Proof Lists

$$t, u ::= \lambda x^A. t \mid x \mid l \quad l ::= \star \mid u :: l$$

Proof Terms and Proof Lists

$$t, u ::= \lambda x^A. t \mid x \mid l \quad l ::= \star \mid u :: l$$

Total Sequents

$$\Gamma \vdash t : A \quad \Gamma; l : A \vdash p$$

Proof Terms and Proof Lists

$$t, u ::= \lambda x^A. t \mid \tilde{x} l \quad l ::= \star \mid u :: l$$

Total Sequents

$$\Gamma \vdash t : A \qquad \Gamma; l : A \vdash p$$

Inference Rules

$$\frac{\Gamma, x : A \vdash t : B}{\Gamma \vdash \lambda x^A. t : A \supset B} \text{ RI}$$

$$\frac{\Gamma, x : A; l : A \vdash p}{\Gamma, x : A \vdash \tilde{x} l : p} \text{ CTR}$$

$$\frac{}{\Gamma; \star : p \vdash p} \text{ AX}$$

$$\frac{\Gamma \vdash u : A \quad \Gamma; l : B \vdash p}{\Gamma; u :: l : A \supset B \vdash p} \text{ LI}$$

Proof Terms and Proof Lists

$$t, u ::= \lambda x^A. t \mid x \mid l \quad l ::= \star \mid u :: l$$

Total Sequents

$$\Gamma \vdash t : A \quad \Gamma; l : A \vdash p$$

Example

$$\begin{array}{c}
 \frac{}{\Gamma; \star : p \vdash p} \text{AX} \\
 \frac{}{\Gamma \vdash x^{\hat{\star}} : p} \text{CTR} \quad \frac{}{\Gamma; \star : p \vdash p} \text{AX} \\
 \frac{}{\Gamma; ((x^{\hat{\star}}) :: \star) : p \supset p \vdash p} \text{LI} \\
 \frac{}{\Gamma \vdash f^{\wedge}((x^{\hat{\star}}) :: \star) : p} \text{CTR} \\
 \hline
 \vdash \lambda f^{(p \supset p)}. \lambda x^p. f^{\wedge}((x^{\hat{\star}}) :: \star) : (p \supset p) \supset p \supset p \quad \text{RI}
 \end{array}$$

where $\Gamma := \{f : p \supset p, x : p\}$

Proof Search (Focusing)

1. Eagerly invert RI to decompose implications in the RHS.

$$\frac{\Gamma, x : A \vdash B}{\Gamma \vdash A \supset B} \text{ } RI$$

Proof Search (Focusing)

1. Eagerly invert RI to decompose implications in the RHS.
2. When the RHS becomes atomic, apply CTR to decide a formula in the LHS to focus on.

$$\frac{\Gamma, x : A \vdash B}{\Gamma \vdash A \supset B} \text{ RI} \qquad \frac{\Gamma, x : A; A \vdash p}{\Gamma, x : A \vdash p} \text{ CTR}$$

Proof Search (Focusing)

1. Eagerly invert RI to decompose implications in the RHS.
2. When the RHS becomes atomic, apply CTR to decide a formula in the LHS to focus on.
3. As long as the focus is an implication, apply LI, keeping the focus on the succedent formula, and launching sub-problems.

$$\frac{\Gamma, x : A \vdash B}{\Gamma \vdash A \supset B} \text{ RI}$$

$$\frac{\Gamma, x : A; A \vdash p}{\Gamma, x : A \vdash p} \text{ CTR}$$

$$\frac{\Gamma \vdash A \quad \Gamma; B \vdash p}{\Gamma; A \supset B \vdash p} \text{ LI}$$

Proof Search (Focusing)

1. Eagerly invert RI to decompose implications in the RHS.
2. When the RHS becomes atomic, apply CTR to decide a formula in the LHS to focus on.
3. As long as the focus is an implication, apply LI, keeping the focus on the succedent formula, and launching sub-problems.
4. When the focus becomes atomic, compare with the atom in the RHS; if equal, finish with AX, otherwise backtrack to 2.

$$\frac{\Gamma, x : A \vdash B}{\Gamma \vdash A \supset B} \text{ RI}$$

$$\frac{\Gamma, x : A; A \vdash p}{\Gamma, x : A \vdash p} \text{ CTR}$$

$$\frac{}{\Gamma; p \vdash p} \text{ AX}$$

$$\frac{\Gamma \vdash A \quad \Gamma; B \vdash p}{\Gamma; A \supset B \vdash p} \text{ LI}$$

- Sequents of LJ_T: $\sigma = (\Gamma \vdash A)$ $\tau = (\Gamma; A \vdash p)$

From LJT to LJT_∂

- Sequents of LJT: $\sigma = (\Gamma \vdash A)$ $\tau = (\Gamma; A \vdash p)$
- Formal Sequents: $\underline{\sigma} = (\underline{\Gamma \vdash A})$ $\underline{\tau} = (\underline{\Gamma; A \vdash p})$

From LJ to LJ_∂

- Sequents of LJ: $\sigma = (\Gamma \vdash A)$ $\tau = (\Gamma; A \vdash p)$
- **Formal Sequents:** $\underline{\sigma} = (\underline{\Gamma \vdash A})$ $\underline{\tau} = (\underline{\Gamma; A \vdash p})$
- **Partial Proof Terms:** proof terms with occurrences of formal sequents;

From LJT to LJT_{∂}

- Sequents of LJT: $\sigma = (\Gamma \vdash A)$ $\tau = (\Gamma; A \vdash p)$
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- **Partial Derivations:** derivations in LJT_{∂} which formalize incomplete proofs found in the proof search process;

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- **Formal Sequents:** $\underline{\sigma} = (\underline{\Gamma} \vdash \underline{A})$ $\underline{\tau} = (\underline{\Gamma}; \underline{A} \vdash \underline{p})$
- **Partial Proof Terms:** proof terms with occurrences of formal sequents;
- **Partial Derivations:** derivations in LJT_{∂} which formalize incomplete proofs found in the proof search process;
- **Rewriting System:** reduction relation on partial proof terms expressing proof search, such that **total terms are normal forms**.

From LJT to LJT_{∂}

- Sequents of LJT: $\sigma = (\Gamma \vdash A)$ $\tau = (\Gamma; A \vdash p)$
- **Formal Sequents:** $\underline{\sigma} = (\underline{\Gamma} \vdash A)$ $\underline{\tau} = (\underline{\Gamma}; A \vdash p)$
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Aim: Proof Search as Normalization

LJT derives $\Gamma \vdash t : A$ iff $\underline{\Gamma} \vdash A \twoheadrightarrow t$

From LJT to LJT_{∂}

- Sequents of LJT: $\sigma = (\Gamma \vdash A)$ $\tau = (\Gamma; A \vdash p)$
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Aim: Proof Search as Normalization

LJT derives $\Gamma \vdash t : A$ iff $\underline{\Gamma} \vdash A \twoheadrightarrow t$
proof search succeeds

From LJT to LJT_{∂}

- Sequents of LJT: $\sigma = (\Gamma \vdash A)$ $\tau = (\Gamma; A \vdash p)$
- **Formal Sequents:** $\underline{\sigma} = (\underline{\Gamma} \vdash A)$ $\underline{\tau} = (\underline{\Gamma}; A \vdash p)$
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Aim: Proof Search as Normalization

LJT derives $\Gamma \vdash t : A$ iff $\underline{\Gamma} \vdash A \twoheadrightarrow t$
proof search succeeds the formal sequent normalizes

Proof Terms and

Proof Lists

Proof Terms
Proof Lists

$$\begin{aligned} t, u &::= \lambda x^A. t \mid x^{\wedge} l \mid \\ l &::= \star \mid u :: l \end{aligned}$$

Partial Proof Terms and Partial Proof Lists

Partial Proof Terms
Partial Proof Lists

$$\begin{aligned} t, u &::= \lambda x^A. t \mid x^{\wedge} l \mid \underline{\sigma} \\ l &::= \star \mid u :: l \mid \underline{\tau} \end{aligned}$$

Partial Proof Terms and Partial Proof Lists

Partial Proof Terms
Partial Proof Lists

$$\begin{aligned} t, u &::= \lambda x^A. t \mid x^{\wedge} l \mid \underline{\sigma} \\ l &::= \star \mid u :: l \mid \underline{\tau} \end{aligned}$$

Formal Term Sequents
Formal List Sequents

$$\begin{aligned} \underline{\sigma} &::= \frac{}{\Gamma \vdash A} \\ \underline{\tau} &::= \frac{}{\Gamma; A \vdash p} \end{aligned}$$

LJT_{∂} : partial proof terms

Partial Proof Terms and Partial Proof Lists

Partial Proof Terms
Partial Proof Lists

$$\begin{aligned} t, u &::= \lambda x^A. t \mid x^* l \mid \underline{\sigma} \\ l &::= \star \mid u :: l \mid \underline{\tau} \end{aligned}$$

Formal Term Sequents
Formal List Sequents

$$\begin{aligned} \underline{\sigma} &::= \frac{}{\Gamma \vdash A} \\ \underline{\tau} &::= \frac{}{\Gamma; A \vdash p} \end{aligned}$$

Examples

Partial Proof Term
Partial Proof List

$$\begin{aligned} &\lambda f^{(p \supset p)}. \lambda x^p. (\underline{\Gamma \vdash p}) \\ (x^* *) &:: (\underline{\Gamma; p \vdash p}) \end{aligned}$$

Partial Sequents

$$\Xi \Vdash t : \sigma$$

$$\Xi \Vdash l : \tau$$

Partial Sequents

Proof State

$\Xi \Vdash t : \sigma$

$\Xi \Vdash l : \tau$

Partial Sequents

Proof State

proof obligations
record of the search
goal sequent

$\Xi \Vdash t : \sigma$

$\Xi \Vdash l : \tau$

list of formal sequents

partial proof term/list

σ/τ

Ξ

t/l

Partial Sequents

Proof State

proof obligations
record of the search
goal sequent

$\Xi \Vdash t : \sigma$

list of formal sequents
partial proof term/list
 σ / τ

$\Xi \Vdash l : \tau$

Ξ
 t/l

Example

Proof State

proof obligations
record of the search
goal sequent

$[\Gamma \vdash p] \Vdash \lambda f^{(p \supset p)}. \lambda x^p. (\Gamma \vdash p) : (\vdash (p \supset p) \supset p \supset p)$
where $\Gamma := \{f : p \supset p, x : p\}$

list of formal sequents
partial proof term/list
 $\vdash (p \supset p) \supset p \supset p$

$[\Gamma \vdash p]$
 $\lambda f^{(p \supset p)}. \lambda x^p. (\Gamma \vdash p)$

Inference Rules

$$\frac{}{[\underline{\sigma}] \Vdash \underline{\sigma} : \sigma} \partial$$

$$\frac{}{\epsilon \Vdash \star : (\Gamma; p \vdash p)} AX$$

$$\frac{\Xi \Vdash t : (\Gamma, x : A \vdash B)}{\Xi \Vdash \lambda x^A. t : (\Gamma \vdash A \supset B)} RI$$

$$\frac{}{[\underline{\tau}] \Vdash \underline{\tau} : \tau} \partial$$

$$\frac{\Xi \Vdash l : (\Gamma, x : A; A \vdash p)}{\Xi \Vdash \hat{x}l : (\Gamma, x : A \vdash p)} CTR$$

$$\frac{\Xi_1 \Vdash u : (\Gamma \vdash A) \quad \Xi_2 \Vdash l : (\Gamma; B \vdash p)}{\Xi_1 @ \Xi_2 \Vdash u :: l : (\Gamma; A \supset B \vdash p)} LI$$

$$\begin{array}{lll}
 (IR) & \frac{\Gamma \vdash A \supset B}{\Gamma \vdash A \supset B} & \rightarrow \lambda x^A. (\Gamma, x : A \vdash B) \\
 (SFL) & \frac{\Gamma, x : A \vdash p}{\Gamma, x : A \vdash p} & \rightarrow x^{\wedge}(\Gamma, x : A; A \vdash p) \\
 (KFL) & \frac{\Gamma; A \supset B \vdash p}{\Gamma; A \supset B \vdash p} & \rightarrow (\underline{\Gamma \vdash A}) :: (\underline{\Gamma; B \vdash p}) \\
 (FFL) & \frac{\Gamma; p \vdash p}{\Gamma; p \vdash p} & \rightarrow \star
 \end{array}$$

$$\begin{array}{lll} (IR) & \frac{\Gamma \vdash A \supset B}{\Gamma \vdash A \supset B} & \rightarrow \lambda x^A. (\Gamma, x : A \vdash B) \\ (SFL) & \frac{\Gamma, x : A \vdash p}{\Gamma, x : A \vdash p} & \rightarrow x^{\wedge}(\Gamma, x : A; A \vdash p) \\ (KFL) & \frac{\Gamma; A \supset B \vdash p}{\Gamma; A \supset B \vdash p} & \rightarrow (\underline{\Gamma \vdash A}) :: (\underline{\Gamma; B \vdash p}) \\ (FFL) & \frac{\Gamma; p \vdash p}{\Gamma; p \vdash p} & \rightarrow \star \end{array}$$

Comments

$$\begin{array}{lll} (IR) & \frac{\Gamma \vdash A \supset B}{\Gamma \vdash A \supset B} & \rightarrow \lambda x^A. (\Gamma, x : A \vdash B) \\ (SFL) & \frac{\Gamma, x : A \vdash p}{\Gamma, x : A \vdash p} & \rightarrow x^{\wedge}(\Gamma, x : A; A \vdash p) \\ (KFL) & \frac{\Gamma; A \supset B \vdash p}{\Gamma; A \supset B \vdash p} & \rightarrow (\underline{\Gamma \vdash A}) :: (\underline{\Gamma; B \vdash p}) \\ (FFL) & \frac{\Gamma; p \vdash p}{\Gamma; p \vdash p} & \rightarrow \star \end{array}$$

Comments

- One reduction rule for each inference rule;

$$\begin{array}{lll}
 (IR) & \frac{\Gamma \vdash A \supset B}{\Gamma \vdash A \supset B} & \rightarrow \lambda x^A. (\Gamma, x : A \vdash B) \\
 (SFL) & \frac{\Gamma, x : A \vdash p}{\Gamma \vdash A \supset p} & \rightarrow x^{\wedge}(\Gamma, x : A; A \vdash p) \\
 (KFL) & \frac{\Gamma; A \supset B \vdash p}{\Gamma \vdash A \supset B} & \rightarrow (\underline{\Gamma \vdash A}) :: (\underline{\Gamma; B \vdash p}) \\
 (FFL) & \frac{\Gamma; p \vdash p}{\Gamma \vdash p} & \rightarrow \star
 \end{array}$$

Comments

- One reduction rule for each inference rule;
- The reduction rule tells how to use the inference rule during proof search;

$$\begin{array}{lll}
 (IR) & \frac{\Gamma \vdash A \supset B}{\Gamma \vdash A \supset B} & \rightarrow \lambda x^A. (\Gamma, x : A \vdash B) \\
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 (KFL) & \frac{\Gamma; A \supset B \vdash p}{\Gamma \vdash A \supset B} & \rightarrow (\Gamma \vdash A) :: (\Gamma; B \vdash p) \\
 (FFL) & \frac{\Gamma; p \vdash p}{\Gamma \vdash p} & \rightarrow \star
 \end{array}$$

Comments

- One reduction rule for each inference rule;
- The reduction rule tells how to use the inference rule during proof search;
- Redexes are formal sequents;

$$\begin{array}{lll}
 (IR) & \frac{\Gamma \vdash A \supset B}{\Gamma \vdash A \supset B} & \rightarrow \lambda x^A. (\Gamma, x : A \vdash B) \\
 (SFL) & \frac{\Gamma, x : A \vdash p}{\Gamma \vdash A \supset p} & \rightarrow x^{\wedge}(\Gamma, x : A; A \vdash p) \\
 (KFL) & \frac{\Gamma; A \supset B \vdash p}{\Gamma \vdash A \supset B} & \rightarrow (\Gamma \vdash A) :: (\Gamma; B \vdash p) \\
 (FFL) & \frac{\Gamma; p \vdash p}{\Gamma \vdash p} & \rightarrow \star
 \end{array}$$

Comments

- One reduction rule for each inference rule;
- The reduction rule tells how to use the inference rule during proof search;
- Redexes are formal sequents;
- Reduction builds the record of the search.

LJT_∂ : example

Informal LJT Derivation

$$\frac{}{\vdash (p \supset p) \supset p \supset p} \partial$$

Reduction Sequence

Proof State

LJT_∂ : example

Informal LJT Derivation

$$\frac{}{\vdash (p \supset p) \supset p \supset p} \partial$$

Reduction Sequence

$$\frac{}{\vdash (p \supset p) \supset p \supset p}$$

Proof State

LJT_∂ : example

Informal LJT Derivation

$$\frac{}{\vdash (p \supset p) \supset p \supset p} \partial$$

Reduction Sequence

$$\underline{\vdash (p \supset p) \supset p \supset p}$$

Proof State

$$[\underline{\vdash (p \supset p) \supset p \supset p}] \Vdash \underline{\vdash (p \supset p) \supset p \supset p} : (\vdash (p \supset p) \supset p \supset p)$$

Informal LJTDerivation

$$\frac{\frac{\overline{\Gamma \vdash p} \quad \partial}{\vdash (p \supset p) \supset p \supset p} \quad RI}{\vdash (p \supset p) \supset p \supset p} \quad RI$$

$$\Gamma := \{f : p \supset p, x : p\}$$

Reduction Sequence

$$\vdash (p \supset p) \supset p \supset p$$

Proof State

LJT_∂ : example

Informal LJTDerivation

$$\frac{\frac{\overline{\Gamma \vdash p} \quad \partial}{\vdash (p \supset p) \supset p \supset p} \quad RI}{\vdash (p \supset p) \supset p \supset p}$$

$$\Gamma := \{f : p \supset p, x : p\}$$

Reduction Sequence

$$\xrightarrow{2}_{IR} \frac{\vdash (p \supset p) \supset p \supset p}{\lambda f (p \supset p) . \lambda x p . (\Gamma \vdash p)}$$

Proof State

LJT_∂ : example

Informal LJTDerivation

$$\frac{\frac{\overline{\Gamma \vdash p} \quad \partial}{\vdash (p \supset p) \supset p \supset p} \quad RI}$$

$$\Gamma := \{f : p \supset p, x : p\}$$

Reduction Sequence

$$\xrightarrow{2}_{IR} \frac{\vdash (p \supset p) \supset p \supset p}{\lambda f (p \supset p) . \lambda x p . (\underline{\Gamma \vdash p})}$$

Proof State

$$[(\underline{\Gamma \vdash p})] \Vdash \lambda f (p \supset p) . \lambda x p . (\underline{\Gamma \vdash p}) : (\vdash (p \supset p) \supset p \supset p)$$

LJT_∂ : example

Informal LJTDerivation

$$\frac{\frac{\frac{\frac{}{\Gamma \vdash p} \partial}{\Gamma; p \vdash p} LI}{\Gamma; p \supset p \vdash p} CTR}{\Gamma \vdash p} RI$$

$$\Gamma := \{f : p \supset p, x : p\}$$

Reduction Sequence

$$\xrightarrow{IR^2} \frac{\vdash (p \supset p) \supset p \supset p}{\lambda f (p \supset p) . \lambda x p . (\Gamma \vdash p)}$$

Proof State

LJT_∂ : example

Informal LJTDerivation

$$\frac{\frac{\frac{\frac{}{\Gamma \vdash p} \partial \quad \frac{}{\Gamma; p \vdash p} \partial}{\Gamma; p \supset p \vdash p} LI}{\Gamma \vdash p} CTR}{\vdash (p \supset p) \supset p \supset p} RI$$

$$\Gamma := \{f : p \supset p, x : p\}$$

Reduction Sequence

$$\begin{aligned} & \vdash (p \supset p) \supset p \supset p \\ \rightarrow_{IR}^2 & \lambda f (p \supset p) . \lambda x p . (\Gamma \vdash p) \\ \rightarrow_{SFL} & \lambda f (p \supset p) . \lambda x p . f'(\Gamma; p \supset p \vdash p) \\ \rightarrow_{KFL} & \lambda f (p \supset p) . \lambda x p . f'((\Gamma \vdash p) :: (\Gamma; p \vdash p)) \end{aligned}$$

Proof State

LJT_∂ : example

Informal LJTDerivation

$$\frac{\frac{\frac{}{\Gamma \vdash p} \partial \quad \frac{}{\Gamma; p \vdash p} \partial}{\Gamma; p \supset p \vdash p} LI}{\Gamma \vdash p} CTR}{\vdash (p \supset p) \supset p \supset p} RI$$

$$\Gamma := \{f : p \supset p, x : p\}$$

Reduction Sequence

$$\begin{aligned} & \vdash (p \supset p) \supset p \supset p \\ \rightarrow_{IR}^2 & \lambda f (p \supset p) . \lambda x^p . (\Gamma \vdash p) \\ \rightarrow_{SFL} & \lambda f (p \supset p) . \lambda x^p . f^*(\Gamma; p \supset p \vdash p) \\ \rightarrow_{KFL} & \lambda f (p \supset p) . \lambda x^p . f^*((\Gamma \vdash p) :: (\Gamma; p \vdash p)) \end{aligned}$$

Proof State

$$[(\Gamma \vdash p), (\Gamma; p \vdash p)] \Vdash \lambda f^{(p \supset p)} . \lambda x^p . f^*((\Gamma \vdash p) :: (\Gamma; p \vdash p)) : (\vdash (p \supset p) \supset p \supset p)$$

Informal LJTDerivation

$$\begin{array}{c}
 \frac{}{\Gamma; p \vdash p} \text{ AX} \\
 \frac{}{\Gamma \vdash p} \text{ CTR} \quad \frac{}{\Gamma; p \vdash p} \text{ AX} \\
 \frac{}{\Gamma \vdash p} \text{ LI} \\
 \frac{}{\Gamma; p \supset p \vdash p} \text{ CTR} \\
 \frac{}{\Gamma \vdash p} \text{ RI} \\
 \frac{}{\vdash (p \supset p) \supset p \supset p} \text{ RI}
 \end{array}$$

$$\Gamma := \{f : p \supset p, x : p\}$$

Reduction Sequence

$$\begin{array}{l}
 \vdash (p \supset p) \supset p \supset p \\
 \xrightarrow{2}_{IR} \lambda f (p \supset p) . \lambda x p . (\Gamma \vdash p) \\
 \xrightarrow{SFL} \lambda f (p \supset p) . \lambda x p . f'(\Gamma; p \supset p \vdash p) \\
 \xrightarrow{KFL} \lambda f (p \supset p) . \lambda x p . f'((\Gamma \vdash p) :: (\Gamma; p \vdash p))
 \end{array}$$

Proof State

LJT_∂ : example

Informal LJTDerivation

$$\begin{array}{c}
 \frac{}{\Gamma; p \vdash p} \text{ AX} \\
 \frac{}{\Gamma \vdash p} \text{ CTR} \quad \frac{}{\Gamma; p \vdash p} \text{ AX} \\
 \frac{}{\Gamma; p \supset p \vdash p} \text{ LI} \\
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 \end{array}$$

$$\Gamma := \{f : p \supset p, x : p\}$$

Reduction Sequence

$$\begin{array}{l}
 \vdash (p \supset p) \supset p \supset p \\
 \rightarrow_{IR}^2 \quad \lambda f (p \supset p) . \lambda x^p . (\Gamma \vdash p) \\
 \rightarrow_{SFL} \quad \lambda f (p \supset p) . \lambda x^p . f^*(\Gamma; p \supset p \vdash p) \\
 \rightarrow_{KFL} \quad \lambda f (p \supset p) . \lambda x^p . f^*((\Gamma \vdash p) :: (\Gamma; p \vdash p)) \\
 \rightarrow_{FFL} \quad \lambda f (p \supset p) . \lambda x^p . f^*((\Gamma \vdash p) :: *) \\
 \rightarrow_{SFL} \quad \lambda f (p \supset p) . \lambda x^p . f^*((x^*(\Gamma; p \vdash p)) :: *) \\
 \rightarrow_{FFL} \quad \lambda f (p \supset p) . \lambda x^p . f^*((x^*) :: *)
 \end{array}$$

Proof State

Informal LJT Derivation

$$\begin{array}{c}
 \frac{}{\Gamma; p \vdash p} \text{ AX} \\
 \frac{}{\Gamma \vdash p} \text{ CTR} \quad \frac{}{\Gamma; p \vdash p} \text{ AX} \\
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$$\Gamma := \{f : p \supset p, x : p\}$$

Reduction Sequence

$$\begin{array}{l}
 \vdash (p \supset p) \supset p \supset p \\
 \xrightarrow{2} \text{IR} \quad \lambda f (p \supset p) . \lambda x^p . (\Gamma \vdash p) \\
 \xrightarrow{\text{SFL}} \lambda f (p \supset p) . \lambda x^p . f^{\sim}(\Gamma; p \supset p \vdash p) \\
 \xrightarrow{\text{KFL}} \lambda f (p \supset p) . \lambda x^p . f^{\sim}((\Gamma \vdash p) :: (\Gamma; p \vdash p)) \\
 \xrightarrow{\text{FFL}} \lambda f (p \supset p) . \lambda x^p . f^{\sim}((\Gamma \vdash p) :: *) \\
 \xrightarrow{\text{SFL}} \lambda f (p \supset p) . \lambda x^p . f^{\sim}((x^{\sim}(\Gamma; p \vdash p)) :: *) \\
 \xrightarrow{\text{FFL}} \lambda f (p \supset p) . \lambda x^p . f^{\sim}((x^{\sim}*) :: *)
 \end{array}$$

Proof State

$$\epsilon \Vdash \lambda f^{(p \supset p)} . \lambda x^p . f^{\sim}((x^{\sim}*) :: *) : (\vdash (p \supset p) \supset p \supset p)$$

Conservativity

- LJT derives $\Gamma \vdash t : A$ iff LJT_{∂} derives $\epsilon \Vdash t : (\Gamma \vdash A)$.
- LJT derives $\Gamma; I : A \vdash p$ iff LJT_{∂} derives $\epsilon \Vdash I : (\Gamma; A \vdash p)$.

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Record of Search

- If $\Xi \Vdash t : \sigma$ then $\underline{\sigma} \twoheadrightarrow t$.
- If $\Xi \Vdash I : \tau$ then $\underline{\tau} \twoheadrightarrow I$.

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Subject Reduction

- If $\Xi \Vdash t : \sigma$ and $t \rightarrow t'$, then there is Ξ' such that $\Xi \rightarrow \Xi'$ and $\Xi' \Vdash t' : \sigma$.
- If $\Xi \Vdash I : \tau$ and $I \rightarrow I'$, then there is Ξ' such that $\Xi \rightarrow \Xi'$ and $\Xi' \Vdash I' : \tau$.

Proof Search as Normalization

- For all $t \in LJT$: LJT derives $\Gamma \vdash t : A$ iff $\underline{\sigma} \twoheadrightarrow t$, where $\underline{\sigma} = \underline{\Gamma \vdash A}$.
- For all $l \in LJT$: LJT derives $\Gamma; l : A \vdash p$ iff $\underline{\tau} \twoheadrightarrow l$, where $\underline{\tau} = \underline{\Gamma; A \vdash p}$.

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Proof Terms - one kind of formal sequent does not constitute, by itself, a partial proof term;

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